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FIELDS Macroeconomics, Fiscal & Monetary Policy, Heterogeneous Agents, Reinforcement Learning

POSITION **University of Copenhagen** 2024 – Present
Postdoctoral Researcher

EDUCATION **Copenhagen Business School** 2020 – 2024
PhD in Economics

Paris School of Economics 2018 – 2020
Master in Analysis and Policy in Economics – *With Honors*

Sciences Po Paris 2014 – 2017
Master in Economics and Public Policy – Cum Laude, *Top 10%*

Sorbonne Paris 1 2014 – 2017
Master in Philosophy and Society – *With Honors*

RESEARCH PROJECTS **Permanent Labor Income Inequality and Stabilization Policies (JMP)**
With Eustache Elina (Banque de France)

We study how rising permanent labor income (PLI) inequality changes the transmission of monetary and fiscal policy. Standard heterogeneous-agent models imply that hand-to-mouth status is unrelated to permanent income, and that the PLI distribution is therefore neutral. In the PSID, instead, the hand-to-mouth share falls from 64% to 22% across PLI quintiles, a gradient that has steepened since 1999. A model with a non-homothetic taste for wealth reproduces these facts and breaks the neutrality. The incidence of fiscal policy becomes a first-order determinant of the output response: the same deficit stimulates or contracts output depending on who receives and who repays it. For monetary policy, the effect of a rate cut depends on how the government rebates its interest savings. The rise in PLI inequality since 1970 also raised the aggregate MPC from 0.30 to 0.36, strengthened tax-financed fiscal packages while weakening transfer-financed ones, and made monetary policy slightly stronger, more regressive, and more dependent on the fiscal rule.

Fiscal Multipliers in Globally Solved HANK Models with Aggregate Risk
With Jeppe Druedahl, Jacob Røpke (University of Copenhagen)

We show that a nonlinear perfect-foresight solution to standard HANK models, computed using standard fast methods, provides a good approximation to the full global solution with aggregate risk, except for a level shift driven by precautionary saving. This result holds in calibrations featuring extensive sign, size, and state dependence in the effects of fiscal policy. We build on, and extend, recent advances in solving

heterogeneous-agent models in moving-average (MA) form using deep learning, and confirm the accuracy of this approach.

Deep Learning Solutions of Large Non-Convex Life-Cycle Models

With Jeppe Druedahl, Jacob Røpke (University of Copenhagen)

We introduce a novel deep learning algorithm for solving large life-cycle models with both continuous and discrete choices. This allows us to simultaneously account for both labor supply choices with human capital accumulation, portfolio choices with a risky and a risk-free asset, and housing and mortgage choices. We work directly on the Bellman equation and approximate both value and policy functions with neural networks, and use a simulated training sample instead of tensor product grids. This substantially alleviates the curse of dimensionality. We demonstrate this in a consumption-saving model with multiple durable goods subject to non-convex adjustment costs where our deep learning algorithm clearly outperforms a standard value function iteration. We confirm that we can accurately solve a large life-cycle model in 12 hours on a single GPU. We solve the model simultaneously across all periods instead of with backward induction. This simplifies transfer learning, where a new solution for a new set of parameters in a calibration or estimation can easily be achieved. An accompanying easy-to-use software package implements the method.

The Exit Channel of Monetary Policy

With Eustache Elina (Banque de France), Léonard Bocquet (Amsterdam)

Monetary tightening can generate inefficient firm exits, by exacerbating firm illiquidity constraints. Contrary to the common belief that only small firms are affected, new evidence shows that large firms are vulnerable too, leading to large aggregate effects. This paper investigates why monetary tightening causes large firms to exit and explores the macroeconomic implications. To do so, we develop a model of endogenous firm exit with financial frictions and partial irreversibility. Financial frictions imply that only productive firms are able to take on debt, making them highly exposed to interest rate changes, as partial irreversibility makes deleveraging costly. The quality of firm selection can therefore endogenously worsen during monetary tightening episodes, highlighting a potential need for support for large firms in distress.

From Income to Wealth Inequality: Trickle-Down vs. Capital Gains

With Eustache Elina (Banque de France)

Higher permanent labor income inequality (PLI) raises saving, since households with higher PLI save a larger share of it. For asset markets to clear, prices must adjust: how does this feed back on wealth inequality? If the saving finances capital, interest rates fall, wages rise, and poorer households gain. If it raises asset prices instead, the capital gains accrue to the rich. In a heterogeneous-agent model with non-homothetic wealth preferences and imperfect competition, markups weaken the first channel and let the second dominate for five decades along the transition. Fed with the observed U.S. rise in inequality since 1970, the model generates two fifths of the rise in the wealth-to-output ratio and a quarter of the rise in Tobin's Q . These price effects raise the average wealth of the top 0.1% by 25% and average wealth by 8% by 2020; under perfect competition both fall, by 4% and 5%.

Why is there still investment in polluting capital? Stranded assets and climate policy uncertainty

Despite governments' commitments to limit global warming to 1.5°C, there is still investment in carbon-intensive capital. This paper uses a growth model featuring irreversible investment, capacity utilisation, clean and polluting capital to study this

apparent paradox. It shows that current investment in polluting capital and CO₂ emissions are coherent with expectations of a future carbon tax, if investors also expect a bailout of polluting capital. This result implies that governments' credibility can play an important role in reducing the cost of implementing an optimal carbon tax by committing not to bail out. However, there exists a temptation for a short-sighted government to boost output and consumption in the short run by announcing a future bailout.

VISITING POSITIONS	Banque de France Visiting Program, <i>France</i> ,	<i>December 2025 - January 2026</i>
	IIES, <i>Sweden</i> , invited by Per Krusell	<i>November - December 2022</i>
	Sciences Po Paris, <i>France</i> , invited by Xavier Ragot,	<i>September - October 2022</i>
CONFERENCES and PRESENTATIONS	Deep Learning Conference, Chapel Hill, <i>USA</i>	<i>September 2026</i>
	Normac, Bergen, <i>Norway</i>	<i>August 2026</i>
	Banque de France,	<i>January 2026</i>
	Danish Central Bank,	<i>November 2025</i>
	Normac, Copenhagen, <i>Denmark</i>	<i>August 2024</i>
	Economix, Paris, <i>France</i> ,	<i>April 2024</i>
	Macro Seminar, Copenhagen University, <i>Denmark</i>	<i>October 2023</i>
	NHH, Bergen, <i>Norway</i>	<i>September 2023</i>
	AFSE, Sciences Po, <i>Paris, France</i>	<i>June 2023</i>
	Macroeconomics and mathematics, BI, <i>Oslo, Norway</i>	<i>December 2022</i>
	Inequality and climate change workshop, CBS, <i>Denmark</i>	<i>November 2022</i>
	Workshop in computational economics, CREST, <i>Paris, France</i>	<i>September 2022</i>
	Macro-group at Statistics Norway, <i>Oslo, Norway</i>	<i>April 2022</i>
	Danish Graduate Program in Economics (flash talk), <i>Denmark</i>	<i>November 2021</i>
	Nordic Initiative for Corporate Economics (discussant), <i>Denmark</i>	<i>September 2021</i>
	Danish Graduate Program in Economics (flash talk), <i>Denmark</i>	<i>November 2020</i>
TEACHING	Lecturer, "Advanced Macroeconomics with Heterogeneous Agents", Graduate, University of Copenhagen	<i>2025-2026</i>
	TA, "Macroeconomics", Undergraduate, CBS	<i>2024</i>
	TA, "Macroeconomics", Undergraduate, CBS	<i>2022</i>
	TA, "Microeconomics", Undergraduate, CBS	<i>2021</i>
	Lecturer, "Mathematics for Economists", Graduate, CBS	<i>2021</i>
	TA, "Econometrics for Firm Data Analysis", Graduate, CBS	<i>2020, 2021</i>
	TA, "Macroeconomics", Undergraduate, <i>Paris-Descartes</i>	<i>2020</i>
LANGUAGES and SOFTWARE	French (native), English (fluent), Spanish (basic)	
	Python, R, Matlab, Dynare, Git/GitHub, L ^A T _E X, Reuters Datastream, Bloomberg	

ACADEMIC SERVICE	Referee for <i>Ecological Economics</i>	2026
	Organizer of the internal KU-macro reading group	(2025)
	Creator and organizer of the Macro Lunches	2023
	Organizer (with Dolores R. Morales) of the DGPE Workshop (100+ participants)	2023
	Main organizer of the Junior Seminar	2022-2023
	Ph.D. representative at the Ph.D. School	2022
PROFESSIONAL EXPERIENCE	OFCE, Paris <i>Research Assistant</i>	<i>September 2019 – July 2020</i>
	CFTC Workers' union, Paris <i>Technical advisor on economic issues</i>	<i>April – August 2018</i>
	HSBC France, Paris <i>Intern Economist</i>	<i>July – December 2017</i>
	Crédit Mutuel-CIC Market Solutions, Paris <i>Intern Economist</i>	<i>December 2016 – July 2017</i>
	Minister of Ecology, Paris <i>Circular Economy Intern</i>	<i>January–July 2015</i>

REFERENCES

1. **David Jinkins**, Supervisor, Associate Professor, Copenhagen Business School, Denmark, dj.eco@cbs.dk, +45 38 15 56 58
2. **Birthe Larsen**, Associate Professor, Copenhagen Business School, Denmark, bl.eco@cbs.dk, +45 61 79 45 25
3. **Jeppe Druedahl**, Associate Professor, University of Copenhagen, Denmark, jeppe.drueahl@econ.ku.dk